

Assignment 10

(Assume all rings are commutative with multiplicative identity)

1. Let X be a metric space and let $p \in X$ be a point. Let $\mathcal{C}(X)$ be the ring of all continuous real-valued continuous functions on X . Show that :
 - a) $\mathfrak{m}_p = \{f \in \mathcal{C}(X) \mid f(p) = 0\}$ defines a maximal ideal in $\mathcal{C}(X)$.
 - b) Show that if X is compact, every maximal ideal is of this type.(This was pending from the previous tutorial)
2. Show that $\mathbb{Z}[X]$ is not a PID.
3. Given two rings R and S , show that every ideal in $R \times S$ is of the form $I \times J$ for some ideals $I \subset R$ and $J \subset S$. Which ones of these are prime ideals ?